Classical Demand I: UMP and EMP

Advanced Microeconomics 1: Economic Decision Making. Lecture 2

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Where we left the question

- **Done, with no assumptions:** household inflation $\pi^h = \sum_l b_l^h \pi_l$ — who faced what.
- **Done, with one assumption:** basket indexation over-compensates — three lines of revealed preference.
- **Missing:** the *right* compensation. “As well off as before” still has no number attached.

Today we cheat, deliberately: assume the household’s ranking is known, and construct the ideal answer. Lectures 3–4 earn back what we assume today.

Today's route

1. A ranking is not yet arithmetic: when does $\succsim$ become a **function** u?
2. Two optimization problems — one natural (UMP), one inverted (EMP).
3. The inverted one produces the cost of living as a mathematical object: $e(p, u)$.
4. From e: the ideal answers to the 2022 question — CV and EV, in euros, per household.

Invoice item 3: define the ideal answer. Every object below exists for that reason.

From ranking to function

Standing assumptions on $\succsim$, and why each is bought

Definition · monotone / locally nonsatiated

$\succsim$ is **monotone** if $y \gg x \Rightarrow y \succ x$; **strongly monotone** if $y \geq x, y \neq x \Rightarrow y \succ x$. It is **locally nonsatiated (LNS)** if for every $x \in X$ and every $\varepsilon > 0$ there is $y \in X$ with $\|y - x\| \leq \varepsilon$ and $y \succ x$.

- Monotone: more is better — plausible for broad aggregates. LNS is weaker: no bliss point nearby.
- Why bought: LNS is exactly what makes households spend their budget — Walras' law, an assumption about data in Lecture 1, becomes a theorem today.

Convexity

Definition · convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is convex: $y \succsim x$ and $z \succsim x$ imply $\alpha y + (1-\alpha)z \succsim x$ for $\alpha \in [0,1]$. **Strictly convex**: the mixture is $\succ x$ whenever $y \neq z$, $\alpha \in (0,1)$.

- Reads as: a taste for diversification; diminishing marginal rates of substitution.
- Why bought: convexity makes the set of optimal choices convex; strict convexity makes demand a function — one bundle per budget, which is what data look like.

Why a utility function can fail to exist

Rationality alone is not enough. The classic counterexample, on $X = \mathbb{R}^2_+$:

Definition · lexicographic preferences

$x \succsim y$ if either $x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$.

- Complete, transitive, strongly monotone, strictly convex — everything so far. Yet no utility function represents it.
- Reason: no two distinct bundles are indifferent, so every indifference set is a single point. A two-dimensional continuum of indifference sets must each receive its own real number, in order — and there are *not enough real numbers*.

Continuity, and Debreu's theorem

Definition · continuity

$\succsim$ is **continuous** if for any sequence $\{(x^n, y^n)\}_{n=1}^{\infty}$ with $x^n \succsim y^n$ for all n, $x^n \to x$ and $y^n \to y$, we have $x \succsim y$. (Preferences cannot jump at the limit.)

- Lexicographic fails it: $x^n = (\frac{1}{n}, 0) \succsim y^n = (0, 1)$ for every n, yet in the limit $y = (0, 1) \succ (0, 0) = x$.

Theorem · Debreu

If the rational preference relation $\succsim$ on X is continuous, there is a *continuous* utility function $u(x)$ that represents $\succsim$.

Proof: beyond our scope; see MWG 3.C. With Lecture 1's easy direction: rational + continuous $\Rightarrow$ representable, and representable $\Rightarrow$ rational.

Ordinal, not cardinal

- u is not unique: any strictly increasing f makes $v = f(u)$ represent the *same* $\succsim$. E.g. $x_1^{\alpha} x_2^{1-\alpha}$ and $\alpha \ln x_1 + (1-\alpha) \ln x_2$: one preference, two functions.
- The dictionary: monotone $\succsim \Rightarrow u$ increasing; (strictly) convex $\succsim \Rightarrow u$ (strictly) **quasiconcave**: $\{y : u(y) \geq u(x)\}$ convex. Both are *ordinal* — they survive increasing transforms.
- Levels of u mean nothing. "Utility fell by 3" is not a sentence. This is precisely why the inflation question cannot stop at u: the answer must be denominated in something real. Euros, by the end of today.

The consumer’s problem

Why an optimization problem at all?

Nothing so far says households *solve* anything. Three reasons to now hypothesise that they do — that observed choices are the *best affordable* ones:

- **The answer we want is itself an optimum.** "The cheapest way to stay as well off as before" is a minimization problem by its very wording. To even *define* the ideal compensation, someone has to optimize — if not the household, then us on its behalf.
- **It is the only bridge from data to welfare.** Without a link between what people *choose* and what makes them *better off*, demand data are mute about the 2022 question. "Chosen = best affordable" is that link — the same load-bearing assumption L1 filed, now given a functional form.
- **It is an as-if hypothesis, not a psychology.** Nobody computes first-order conditions in a supermarket. The claim is that choices are *consistent enough* to be described this way — and the practical tests exactly that, choice by choice, on you.

The utility maximization problem

Rational, continuous, LNS $\succsim$ on $X = \mathbb{R}^L_+$, represented by continuous u; prices $p \gg 0$, wealth $w > 0$:

$$\text{(UMP)} \qquad \max_{x \geq 0} \ u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

Theorem · existence

If $u(\cdot)$ is continuous, the UMP has a solution. Proof

Two objects fall out, and the rest of the block lives off them: the **maximizer** $x(p, w)$ and the **value** $v(p, w)$.

Walrasian demand: Lecture 1's laws become theorems

Theorem · MWG 3.D.2

If u is a continuous utility function representing an LNS preference relation $\succsim$ on $\mathbb{R}^L_+$, then $x(p, w)$ satisfies

(i) **homogeneity of degree zero**: $x(\alpha p, \alpha w) = x(p, w)$ for $\alpha > 0$;

(ii) **Walras' law**: $p \cdot x = w$ for all $x \in x(p, w)$;

(iii) **convexity/uniqueness**: convex $\succsim \Rightarrow x(p, w)$ a convex set; strictly convex $\succsim \Rightarrow$ single-valued.

Proof

In Lecture 1, (i) and (ii) were *assumptions* about data. Under today's hypothesis they are consequences — the first repayment of the loan.

Solving it: Kuhn–Tucker

If u is continuously differentiable, an optimum x^* admits a multiplier $\lambda \geq 0$ with, for each ℓ:

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell, \qquad \text{with equality if } x_\ell^* > 0.$$

- Interior optimum ($x^* \gg 0$): for any two goods, $\dfrac{\partial u(x^*)/\partial x_\ell}{\partial u(x^*)/\partial x_k} = \dfrac{p_\ell}{p_k}$ — **MRS** $=$ **price ratio**. Only relative prices appear: h.d.0, visible in the first-order condition.
- λ is the shadow value of the budget: $\nabla u \cdot D_w x = \lambda\, p \cdot D_w x = \lambda$ — the marginal utility of one more euro.
- The conditions are also sufficient if u is quasiconcave and monotone with $\nabla u(x) \neq 0$.

Worked: Cobb–Douglas

$u(x) = x_1^{\alpha} x_2^{1-\alpha}$, $\alpha \in (0,1)$. Interior FOCs + Walras' law give

$$x_1(p,w) = \frac{\alpha w}{p_1}, \qquad x_2(p,w) = \frac{(1-\alpha)\,w}{p_2}.$$

- **Constant budget shares**: $b_1 = \alpha$, $b_2 = 1-\alpha$ — exactly the b_l^h of the motivation charts. A Cobb–Douglas household *is* a row of the NBB quartile table.
- Small enough to carry through the block: the ideal answer is computed for this household before the hour ends.
- EC1: the subsistence version $u = (x_1 - s_1)^{\alpha}(x_2 - s_2)^{1-\alpha}$ — shares no longer constant; the poor pinned to necessities.

The indirect utility function

Definition · indirect utility

$v(p, w) = u(x^*)$ for any $x^* \in x(p, w)$: the maximal utility attainable at (p, w).

Proposition · properties (MWG 3.D.3)

$v(p, w)$ is (i) h.d.0; (ii) increasing in w and nonincreasing in p_ℓ; (iii) quasiconvex: $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex; (iv) continuous.

Proof deferred to MWG. A fifth property — Roy's identity — is Lecture 3's engine.

"Worse off from the price change" now has a formal reading: $v(p^1, w) < v(p^0, w)$. A *direction*, at last — but not yet a *size*.

Why v cannot be the final answer

- v depends on which representation we picked: replace u by $f(u)$ and every value of v changes. The size of $v(p^1, w) - v(p^0, w)$ is meaningless.

- Belgium does not index wages in utils.

- The fix is to change the question. Instead of "how much utility from this budget?", invert: "how much budget for this standard of living?" Money in, money out.

The answer’s object: the expenditure function

The expenditure minimization problem

For $p \gg 0$ and $u > u(0)$:

$$\text{(EMP)} \qquad \min_{x \geq 0} \; p \cdot x \quad \text{s.t.} \quad u(x) \geq u.$$

Definition · expenditure function

$e(p, u) = p \cdot x^*$ for any solution x^*: the *minimal cost of reaching the standard of living u at prices p*.

- Read it again with 2022 in mind: $e(p^{2022}, u^{2021})$ is **what it costs to be as well off as before** — the phrase the course has been chasing, now a mathematical object.
- The *dual* of the UMP: objective and constraint swap roles.

UMP and EMP are the same problem, run backwards

Theorem · MWG 3.E.1

Under the maintained assumptions (continuity, LNS), for $p \gg 0$:

(i) x^* optimal in the UMP at wealth $w > 0 \Rightarrow x^*$ optimal in the EMP at required utility $u(x^*)$; the minimized expenditure is exactly w.

(ii) x^* optimal in the EMP at $u > u(0) \Rightarrow x^*$ optimal in the UMP at wealth $p \cdot x^*$; the maximized utility is exactly u.

Proof

Hence the two value functions are inverses of each other in their last argument:

$$e(p, v(p, w)) \equiv w, \qquad v(p, e(p, u)) \equiv u.$$

Properties of the cost of living

Theorem · MWG 3.E.2

$e(p, u)$ is

(i) homogeneous of degree **one** in p;

(ii) strictly increasing in u and nondecreasing in each p_ℓ;

(iii) **concave** in p;

(iv) continuous in p, for $p \gg 0$.

Proof

- (i): double every price and the cost of the old life doubles — the indexation instinct, exactly right *for proportional* inflation. 2022 was not proportional.
- (iii) is the sleeper. Concavity: as one price rises, cost rises *less than linearly*, because the household substitutes. Lecture 1's over-compensation theorem is the shadow of this curvature. Lecture 3 measures it.

Hicksian demand

Definition · Hicksian (compensated) demand

$h(p,u) \subseteq \mathbb{R}^L_+$: the solution set of the EMP — the cheapest bundle(s) delivering the standard of living u at prices p.

Theorem · MWG 3.E.3

For $p \gg 0$: (i) **h.d.0 in** p: $h(\alpha p, u) = h(p,u)$; (ii) **no excess utility**: $u(x) = u$ for any $x \in h(p,u)$; (iii) convex $\succsim \Rightarrow h(p,u)$ convex; strictly convex $\Rightarrow$ single-valued. Proof

Duality identities, from the equivalence theorem:

$$h(p,u) \equiv x\big(p, e(p,u)\big), \qquad x(p,w) \equiv h\big(p, v(p,w)\big).$$

Not directly observable — no dataset holds utility fixed. Its road to the data is Lecture 3.

Two compensations

From (p^0, w) with $x^0 = x(p^0, w)$, prices move to p^1. Two ways to “compensate”:

Slutsky (Lecture 1)

$\Delta w_S = p^1 \cdot x^0 - w$: keep the *old bundle* affordable.

Observable today. **Over-pays** (the L1 theorem). This is what basket indexation does.

Hicksian (today)

$\Delta w_H = e(p^1, u^0) - w$: keep the *old standard of living* reachable.

Exact — the ideal answer. Needs preferences.

$\Delta w_H \;\leq\; \Delta w_S$ since $e(p^1, u^0) \leq p^1 \cdot x^0$ (x^0 attains u^0, so it is feasible in the EMP).

The gap between the columns is the substitution bias — the euro amount at stake in the indexation debate.

The compensated law of demand, from preferences

Theorem · MWG 3.E.4

Under the maintained assumptions, with $h(p,u)$ single-valued for $p \gg 0$: for all p' and p'',

$$(p'' - p') \cdot \left[h(p'', u) - h(p', u)\right] \;\le\; 0.$$

Proof

- Own-price version: compensated own-price effects are nonpositive, globally. *Marshallian* demand obeys no such law (Giffen); holding the standard of living fixed is what buys the sign.
- Lecture 1 proved this law from WARP, for *Slutsky*-compensated changes. Same law, second derivation, deeper hypothesis. The two constructions collide — productively — in Lecture 3.

The ideal answer, stated

Money-metric utility: euros as the unit

Fix any reference prices $\bar{p} \gg 0$ and consider, as a function of (p, w):

$$e\big(\bar{p},\ v(p, w)\big).$$

- This is itself an indirect utility function — a strictly increasing transform of v — but its unit is **euros at prices** $\bar{p}$: the budget needed, at reference prices, to live as well as you do at (p, w).
- Differences of *this* object are meaningful money amounts. The representation-dependence of v is gone.
- One free choice remains: *which* $\bar{p}$. The two natural candidates are the old prices and the new — and each choice has a name.

The ideal answers: EV and CV

Let $u^0 = v(p^0, w)$, $u^1 = v(p^1, w)$; note $e(p^0, u^0) = e(p^1, u^1) = w$.

Definition · equivalent variation

$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0) = e(p^0, u^1) - w$: the euro amount, *at old prices*, equivalent to the price change.

Definition · compensating variation

$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0) = w - e(p^1, u^0)$: minus the payment that would exactly restore the old standard of living *at new prices*.

- For 2022: $-CV^h = e^h(p^1, u^0) - w$ is the correct indexation payment for household h — the number the health index approximates with one basket and one average household.
- Both are ideal; they differ in the reference. Which one policy *owes* is a genuine choice — it returns, with teeth, in Lecture 4.

The true index vs. the published index

Definition · Konüs (true cost-of-living) index

$$P_K(p^0 \to p^1; u^0) = \frac{e(p^1, u^0)}{e(p^0, u^0)} \qquad \text{vs. Laspeyres:} \qquad P_L = \frac{p^1 \cdot x^0}{p^0 \cdot x^0}.$$

Proposition · Laspeyres bounds the truth from above

$P_K \leq P_L$. *Proof:* x^0 attains u^0, so $e(p^1, u^0) \leq p^1 \cdot x^0$; and $e(p^0, u^0) = p^0 \cdot x^0 = w$. □

- Two lines — Lecture 1's three-line theorem, rewritten in the answer's own language. Fixed-basket indices (the CPI family, the health index) are Laspeyres-type: they mechanically over-state the rise in the cost of living.
- By how much? A second-order question — literally. Lecture 3.

Both objects, computed on UK scanner data through the 2021–23 surge: Chen, Levell & O'Connell (2025) — their "quality-of-living index" is exactly the proportional CV of the last slide. §3 is this lecture's notation in the field; Lectures 3–4 use their numbers.

The ideal answer, computed

Cobb–Douglas household, goods $=$ (energy, other), share α on energy. Then

$$e(p,u) = u\left(\frac{p_E}{\alpha}\right)^{\alpha}\left(\frac{p_O}{1-\alpha}\right)^{1-\alpha} \quad\Longrightarrow\quad \frac{e(p^1,u^0)}{w} = \left(\frac{p_E^1}{p_E^0}\right)^{\alpha} \quad \text{(only } p_E \text{ moves).}$$

Energy prices rise 50%. Two households from the motivation chart:

	exact: $-CV/w = 1.5^{\alpha} - 1$	basket: $\Delta w_S/w = \alpha \times 50\%$	over-pay
poor, $\alpha = 0.10$	4.1%	5.0%	0.9 pp
rich, $\alpha = 0.06$	2.5%	3.0%	0.5 pp

Both L1 results, now with numbers: the *right* payments differ across households (4.1 vs 2.5), and the basket over-pays both. One index, one number, misses both facts.

Where the question stands

Takeaways

- **The ideal answer exists**: $e(p^1, u^0) - w$, per household — "as well off as before" is now arithmetic. EV and CV name the two natural references.
- **Two accounting laws upgraded**: h.d.0 and Walras' law are theorems under the maintained hypothesis, not assumptions about data.
- **The catch, stated plainly**: everything today was built from u — which no dataset contains. The answer is ideal and, so far, *uncomputable*.
- **Next**: Lecture 3 connects e, h and v to observable demand. Four bridges; all of them derivatives.

Required: MWG 3.A–3.E.
Papers: Hausman (AER 1981), §I–II, before Lecture 4; Chen–Levell–O'Connell (IFS 2025), §3 — tonight's notation, in the field.

Ahead: Lecture 3 on Monday; the practical follows Lecture 4, on Monday 5 October.

Appendix: proofs

Proof: the UMP has a solution

- The budget set $B_{p,w} = \{x \in \mathbb{R}^L_+ : p \cdot x \leq w\}$ is **bounded**: for each ℓ, $x_\ell \leq w/p_\ell$ for every $x \in B_{p,w}$ (this uses $p \gg 0$).
- It is **closed**: an intersection of closed half-spaces with the closed orthant.
- Closed and bounded in $\mathbb{R}^L$ = **compact**; and $B_{p,w} \neq \emptyset$ since $0 \in B_{p,w}$. A continuous function attains a maximum on a nonempty compact set. ■

Where it can fail: some $p_\ell = 0$ (budget set unbounded), or u discontinuous — lexicographic preferences again. Another job continuity does.

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Proof: properties of Walrasian demand (MWG 3.D.2)

- *(i) h.d.0.* $B_{\alpha p,\alpha w} = \{x : \alpha p \cdot x \leq \alpha w\} = B_{p,w}$: same constraint set, same objective, hence the same maximizers.

- *(ii) Walras' law.* Suppose $x \in x(p, w)$ with $p \cdot x < w$. Then a small ball around x lies inside the budget set, and LNS supplies y in that ball with $y \succ x$: an affordable strict improvement, contradicting optimality. So $p \cdot x = w$.

- *(iii) Convexity/uniqueness.* Let $x, x' \in x(p, w)$, both with utility u^*. The mixture $x'' = \alpha x + (1 - \alpha)x'$ is affordable ($B_{p,w}$ convex) and quasiconcavity gives $u(x'') \geq u^*$, so $x'' \in x(p, w)$: the solution set is convex. If u is *strictly* quasiconcave and $x \neq x'$, then $u(x'') > u^*$ — contradicting maximality of u^*. So the solution is unique. ∎

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Proof: UMP $\Longleftrightarrow$ EMP (MWG 3.E.1)

(i) Let x^* solve the UMP at wealth w; by Walras' law, $p \cdot x^* = w$. Suppose x^* does *not* solve the EMP at utility $u(x^*)$: some x' has $u(x') \geq u(x^*)$ and $p \cdot x' < w$. By LNS there is x'' near x' with $u(x'') > u(x') \geq u(x^*)$ and still $p \cdot x'' < w$ — an affordable bundle beating x^* in the UMP. Contradiction. The minimized expenditure is $p \cdot x^* = w$.

(ii) Let x^* solve the EMP at $u > u(0)$; then $x^* \neq 0$, so $w' := p \cdot x^* > 0$. Suppose x^* does *not* solve the UMP at w': some x' has $p \cdot x' \leq w'$ and $u(x') > u(x^*) \geq u$. By continuity, $\alpha x'$ with $\alpha < 1$ close to 1 still has $u(\alpha x') > u$, at cost $\alpha\, p \cdot x' < w'$ — cheaper and feasible in the EMP, contradicting minimality. Finally, $u(x^*) = u$ exactly: if $u(x^*) > u$, the same scaling applied to x^* itself yields a cheaper feasible bundle. ■

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Proof: properties of the expenditure function (MWG 3.E.2)

- *(i) h.d.1.* The constraint set $\{x : u(x) \geq u\}$ does not involve p; scaling $p \to \alpha p$ scales the objective, hence the minimized value, by α.
- *(ii) Strictly increasing in u.* Suppose $u' > u$ but $e(p, u') \leq e(p, u)$, and let x' attain u' at cost $e(p, u')$. By continuity, $\alpha x'$ with $\alpha < 1$ near 1 still gives utility $> u$, at cost $\alpha\, e(p, u') < e(p, u)$: contradicts the definition of $e(p, u)$. (Nondecreasing in p_ℓ: raising p_ℓ weakly raises the cost of every feasible bundle.)
- *(iii) Concavity.* Fix $\bar{u}$; let $p'' = \alpha p + (1-\alpha)p'$ and let x'' be optimal at p''. Then x'' attains $\bar{u}$, hence is *feasible* (not necessarily optimal) at prices p and at prices p':

$$e(p'', \bar{u}) = p'' \cdot x'' = \alpha\, p \cdot x'' + (1-\alpha)\, p' \cdot x'' \;\geq\; \alpha\, e(p, \bar{u}) + (1-\alpha)\, e(p', \bar{u}). \quad \blacksquare$$

(iv) Continuity: the maximum theorem; see MWG.

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Proof: properties of Hicksian demand (MWG 3.E.3)

- *(i) h.d.0 in p.* Minimizing $\alpha p \cdot x$ and minimizing $p \cdot x$ over the same constraint set $\{x : u(x) \geq u\}$ have the same solutions.

- *(ii) No excess utility.* Suppose $x \in h(p, u)$ with $u(x) > u$. Since $u > u(0)$, $x \neq 0$, so $p \cdot x > 0$. By continuity, αx with $\alpha < 1$ near 1 still has $u(\alpha x) > u$ — feasible and strictly cheaper. Contradiction; so $u(x) = u$.

- *(iii) Convexity/uniqueness.* The constraint set $\{x : u(x) \geq u\}$ is convex when u is quasiconcave, and the objective is linear: two minimizers cost the same, and any mixture of them is feasible and costs the same — the solution set is convex. Under *strict* convexity, the mixture of two distinct minimizers lies strictly inside the upper contour set, so a small radial contraction stays feasible and cuts cost — forcing uniqueness. ■

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Proof: the compensated law of demand (MWG 3.E.4)

Fix u. Each of $h(p', u)$ and $h(p'', u)$ attains utility u, so each is *feasible* in the other's problem. Cost-minimality therefore gives two inequalities:

$$\begin{aligned} p'' \cdot h(p'', u) &\leq p'' \cdot h(p', u) \qquad (h(p', u) \text{ feasible at prices } p''), \\ p' \cdot h(p', u) &\leq p' \cdot h(p'', u) \qquad (h(p'', u) \text{ feasible at prices } p'). \end{aligned}$$

Add the first to minus the second:

$$(p'' - p') \cdot \left[h(p'', u) - h(p', u)\right] \leq 0. \blacksquare$$

Two lines of feasibility — no calculus, no WARP. Compare Lecture 1: the same law from choice consistency needed the compensation construction. From preferences it is nearly free.

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